Problem 1. (5+5+4+4+2 = 20 points) Let $A = (2, -5, 6)$, $B = (3, -5, 7)$, and $C = (3, -6, 8)$.

(a) Compute $\vec{AB} \times \vec{AC}$ and $\vec{AB} \cdot \vec{AC}$.

$\vec{AB} = \langle 1, 0, 1 \rangle$

$\vec{AC} = \langle 1, -1, 2 \rangle$

$\vec{AB} \times \vec{AC} = \langle 1, -1, -1 \rangle$

$\vec{AB} \cdot \vec{AC} = 3$

Answers: (a) $\langle 1, -1, -1 \rangle$ 3

(b) Find the angle between the vectors $\vec{AC}$ and $\vec{AB}$.

Call the angle θ.

$$\theta = \arccos\left(\frac{\vec{AB} \cdot \vec{AC}}{|\vec{AB}| \cdot |\vec{AC}|}\right) = \arccos\left(\frac{3}{\sqrt{2}\sqrt{6}}\right)$$

$$= \arccos\left(\frac{\sqrt{3}}{2}\right) = \pi/6$$

Answer: (b) $\pi/6$

(c) Find a parametric equation for the line which contains the point B and is parallel to the vector $\vec{AC}$

$$\vec{r}(s) = \langle 3+s, -5-s, 7+2s \rangle$$

Answer: (c)

(d) Find the equation for the plane containing the points A, B, and C.

The plane of interest contains $(2,-5,6)$ and is orthogonal to $\vec{AB} \times \vec{AC}$. Use standard form.

Answer: (d) $x - y - z - 1 = 0$

(e) True or False: The line defined in part (c) must lie in the plane defined in part (d). Justify your answer.

We check:

$$(3+s) - (-5-s) - (7+2s) - 1 \stackrel{?}{=} 0$$

Yes.

Answer: (e) True.

Problem 2. (5 + 5 = 10 points) Consider the curve $\mathcal{C}$ parametrized by

$$\mathrm{r}(\theta) = \langle \sin^3(\theta), \cos^3(\theta), \sin^2(\theta) \rangle$$

for $0 \leq \theta \leq \pi/2$.

(a) Provide a parametric equation for the line tangent to $\mathcal{C}$ at the point $(\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, \frac{1}{2})$.

Note: $\vec{r}(\pi/4) = \langle 1/2\sqrt{2}, 1/2\sqrt{2}, 1/2 \rangle$. We need to calculate $\vec{r}\,'(\pi/4)$.

Since $\vec{r}\,'(\theta) = \langle 3\sin^2(\theta)\cos(\theta), -3\cos^2(\theta)\sin(\theta), 2\sin(\theta)\cos(\theta)$

we have $\vec{r}\,'(\pi/4) = \langle 3/2\sqrt{2}, -3/2\sqrt{2}, 1 \rangle$ and so

$$\ell(t) = \left\langle \frac{1}{2\sqrt{2}} + \frac{3t}{2\sqrt{2}}, \frac{1}{2\sqrt{2}} - \frac{3t}{2\sqrt{2}}, \frac{1}{2} + t \right\rangle$$

Answer: (a) $\frac{1}{2\sqrt{2}} \langle 1+3t, 1-3t, \sqrt{2} + 2\sqrt{2}\,t \rangle$

(b) Compute the length of $\mathcal{C}$.

The length of $\mathcal{C}$ is:

$$\int_0^{\pi/2} |\vec{r}\,'(\theta)|\, d\theta = \int_0^{\pi/2} \sqrt{9\sin^4(\theta)\cos^2(\theta) + 9\cos^4(\theta)\sin^2(\theta) + 4\sin^2(\theta)\cos^2(\theta)}$$

$$= \int_0^{\pi/2} \sqrt{13}\, \sin(\theta)\cos(\theta)\, d\theta$$

$$= \sqrt{13} \left. \frac{\sin^2(\theta)}{2} \right|_0^{\pi/2} = \frac{\sqrt{13}}{2}$$

Answer: (b) $\sqrt{13}/2$

Problem 3. $(5+5+5+5=20$ points) Consider the function $f(x,y)=y^2-x^2-yx^3$.
(a) Compute $\nabla f(1,-1)$.

Answer: | (a) $\nabla f(1,-1) = \langle 1,-3\rangle$ |

(b) Find an equation for the tangent plane to the graph of $z=f(x,y)$ at the point $(1,-1,1)$.

The tangent plane is perpendicular to the gradient of the equation $0=f(x,y)-z$. at the point $(1,-1,1)$. Hence we use

$\vec{n} = \langle 1,-3,-1\rangle$

point : $(1,-1,1)$ $\rightsquigarrow$ $x-3y-z-3=0$

Answer: | (b) $x-3y-z-3=0.$ |

(c) Compute the directional derivative of f at the point $(1,-1)$ in the direction of $\langle 3,1\rangle$.

$$(D_{\vec{u}} f)(-1,1) = \nabla f(-1,1) \cdot \frac{\langle 3,1\rangle}{\sqrt{10}} = 0.$$

Answer: (c) 0

(d) True or false: The level curve defined by $f(x,y) = 1$ has, at the point $(1,-1)$, tangent line parallel to $\langle 3,1\rangle$. Justify your answer.

Since $\nabla f(1,-1) \cdot \langle 3,1\rangle = 0$, we conclude that $\langle 3,1\rangle$ is perpendicular to $\nabla f(1,-1)$. Hence, as we are in 2-D, $\langle 3,1\rangle$ is parallel to the tangent to the level curve $f(x,y) = 1$ at the point $(1,-1)$.

Answer: (d) True

Problem 4. (10+5+5 =20 points) Throughout this problem $f(x,y) = x^2 + y^2 - xy$.
(a) Let S^1 denote the unit circle in the plane, that is, $S^1 = \{(x,y) \mid x^2 + y^2 = 1\}$. Use the method of Lagrange multipliers to find the maximum and minimum values obtained by $f(x,y)$ on S^1.

Since we are restricting to the curve $x^2+y^2=1$, we have

$$f(x,y) = 1-xy$$
$$g(x,y) = x^2+y^2 = 1.$$

Using Lagrange multipliers, we want to solve:

- $-y = \lambda 2x$
- $-x = \lambda 2y$
- $x^2+y^2=1.$

Note: Neither x or y can be 0 as that would imply $x=y=0$ and so $0^2+0^2=1$.

Thus, $\lambda = \frac{-y}{2x} = \frac{-x}{2y} \Rightarrow y^2 = x^2$. So

$$2x^2 = 1 \Rightarrow x = \pm 1/\sqrt{2}$$
$$y = \pm 1/\sqrt{2}$$

So

points of interest:	value of f
$(\sqrt{2}/2, \sqrt{2}/2)$	$1/2$
$(\sqrt{2}/2, -\sqrt{2}/2)$	$3/2$
$(-\sqrt{2}/2, \sqrt{2}/2)$	$3/2$
$(-\sqrt{2}/2, -\sqrt{2}/2)$	$1/2$

Answer: (a) $1/2,\ 3/2$

(b) Find the critical point of $f(x, y)$.

We want to solve

$$\nabla f = \vec{0}$$

$$\Downarrow$$

$$\langle 2x - y, 2y - x \rangle = \langle 0, 0 \rangle$$

$$\Rightarrow y = 2x \quad x = 2y$$

$$\Rightarrow y = 4y \Rightarrow y = 0, x = 0$$

Answer: (b) $(0,0)$

(c) Let D denote the unit disk; that is, $D = \{(x, y) \mid x^2 + y^2 \leq 1\}$. Combine parts (a) and (c) to determine the maximum and minimum values for $f(x, y)$ on D.

(x,y)	$f(x,y)$
$(0,0)$	0
$(\sqrt{2}/2, \sqrt{2}/2)$	$1/2$
$(\sqrt{2}/2, -\sqrt{2}/2)$	$3/2$

Answer: (c) $0, \ 3/2.$

Problem 5. (5+7+8=20 points) Throughout this problem $f(x, y) = x^3 + y^2 - 6xy + 6x + 3y$.
(a) Compute ∇f at an arbitrary point (x, y).

Answer: (a) $\nabla f(x, y) = \langle 3x^2 - 6y + 6, 2y - 6x + 3 \rangle$

(b) Find all critical points of $f(x, y)$.

$\nabla f(x,y) = \vec{0}$

$\Rightarrow 2y = 6x - 3$ and $3x^2 - 6y + 6 = 0$

$\Rightarrow 3x^2 - 18x + 15 = 0$

$\Rightarrow (3x - 3)\overset{(x-5)}{\cancel{(x-15)}} = 0$

$\Rightarrow x = 1$ or $x = 5$

Answer: (b) $(1, 3/2)$, $(5, 27/2)$

(c) Use the second derivatives test to classify the critical points found in part (b).

$f_{xx}(x,y) = 6x$

$f_{yy}(x,y) = 2$

$f_{xy} = -6$

$D = (6x)(2) - (-6)^2 = 12x - 36$
$= 12(x-3)$

For $(1, 3/2)$: $D < 0 \Rightarrow$ saddle point

For $(5, 27/2)$: $D > 0$ and $f_{xx} > 0$ so local min.

Answer: (c) D $(1, 3/2)$: saddle
$(5, 27/2)$: local min here

Problem 6. (2x5=10 points) Let $f(x, y) = e^{x-y}$.
(a) Suppose x and y are functions of t with $x(3) = y(3) = 1/2$ and $x'(3) = 2$ while $y'(3) = 4$. Compute $\frac{df}{dt}(3)$.

$$f_t = f_x x_t + f_y y_t$$

$$\Rightarrow f_t(3) = f_x(1/2, 1/2)x'(3) + f_y(1/2, 1/2)\, y'(3)$$

$$= 1 \cdot 2 + (-1)(4) = \boxed{-2}$$

Note: $f_x = e^{x-y}$

$f_y = -e^{x-y}$

Answer: (a) -2

(b) Suppose x and y are functions of s and t with $x(3,2) = 7$, $y(3,2) = -5$, $x_s(3,2) = 8$, $x_t(3,2) = 1$, $y_s(3,2) = 5$ and $y_t(3,2) = -1$. Compute the partial of f with respect to t at the point $(3,2)$ (that is, the point where $s = 3$ and $t = 2$).

$$f_t = f_x x_t + f_y y_t$$

$$\Rightarrow f_t(3,2) = e^{7-(-5)}(1) + (-e^{7-(-5)})(-1)$$

$$= e^{12} - e^{12}(-1) = \boxed{2e^{12}}$$

Answer: (b) $2e^{12}$